



Can Solar Cells Provide Both Power And Drinking Water?

Energy shortage and clean water scarcity are two key challenges for global sustainable development. The concept of simultaneous production of clean water and electricity by solar energy has been recently investigated by several groups



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The world is on the verge of a water crisis. Rainfall shifts caused by climate change plus escalating water demands of a growing world population threaten society's ability to meet its mounting needs. By 2025, the United Nations (UN) predicts that 2.4 billion people will live in regions of intense water scarcity, which may force as many as 700 million people to travel a long distance from their home in search of water by 2030.

These water woes have made people thirstily eye the more than one sextillion litres of water in Earth's oceans, saline lakes and some underground aquifers with high salt content as a source of drinking water after desalination. (Desalination is necessary to make saline water fit for drinking or irrigation.)

India has sixteen per cent of the total world population, but only four per cent water resources. Seventy per cent of

Rajasthan is arid, and 31 per cent of its groundwater and many lakes are saline. It is estimated that sixty per cent of the area in Rann of Kutch in Gujarat contains native saline water. And there is extreme water scarcity in Chennai due to negligence of water governance. These are just a few places that require installation of desalination plants urgently.

Technology overview

A device that can produce electricity from sunlight while simultaneously purifying water from sea/polluted water has been invented by researchers. This invention can solve two problems in one stroke—the device is not only a source of green energy but also offers a cheaper alternative to current technologies for purifying water.

Conventional technologies often consume large amounts of electricity and require infrastructure beyond the reach of many communities that lack basic access to safe drinking water—a situation thought to affect more than 780 million people worldwide. These people, it has been estimated, spend a collective 200 million hours each day to fetch water from distant sources.

With solar farms often located in arid regions, the new device could provide clean water where it is needed the most. It could even be installed by an individual in the backyard. By having significant amount of fresh water produced continuously on daily basis many challenging tasks can be easily achieved.

The generated clean water can be used for drinking, farming, cleaning solar panels and making desert farming possible, while waste water after cleaning can be used for growing grass below the solar panels to



Fig. 1: Artist's impression of dual-purpose solar cells